



Research paper

Refraction-based design of a butterfly-shaped waveguide for coastal wave sheltering

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A B S T R A C T

A novel waveguide design based on wave refraction principles is proposed, featuring a symmetrical ‘butterfly shape’ with a larger internal water depth and a front edge angled relative to the incident wave direction. By adjusting the front edge angle and local water depth, this structure effectively redirects wave propagation, significantly reducing downstream wave heights. A theoretical design framework is developed and validated against a non-hydrostatic water wave model. The influence of the internal-to-external water-depth ratio, as well as the angle between the waveguide front incident wave ($\theta = \pi/10 - \pi/4$) on refraction patterns was investigated. Results indicate the waveguide effectively reduces wave heights in a defined shelter region, achieving a maximum attenuation exceeding 90% in specific scenarios, while creating a fan-shaped amplification area along the shelter's edge. Simulations with varying incident parameters reveal that sheltering performance depends on both wave period and height. Specifically, wave periods modify the downstream shelter region by altering the refraction length, whereas wave height affects sheltering via diffraction at the waveguide's lee. For an economical design, a depth ratio of $h_2/h_1 = 1.3$ is optimal. The front-wing angle is period-dependent, requiring smaller angles for long-period waves and larger angles for short-period waves.

1. Introduction

The use of engineered structures to attenuate wave energy commonly includes rubble mound breakwaters, vertical breakwaters, and floating breakwaters, amongst others. Given that rock breakwaters have long construction and decommissioning timeframes, they are not suitable for providing temporary wave protection. Floating breakwaters, while often used for temporary wave protection in such scenarios, may be ineffective when the resonance frequency of long-period waves coincides with the natural frequency of the breakwater, rendering them less effective at attenuating medium-to long-period waves (Ji et al., 2018; Gao et al., 2023). When waves interact with submerged seabed features, such as trenches or ridges, their amplitude can be altered due to changes in water depth (Miles, 1982). If wave amplitude can be controlled by modifying the water depth, it is possible to reduce wave height in specific regions solely through underwater dredging operations, without any land-based operations, ensuring that both construction and maritime activities proceed smoothly. Wave interactions with trenches have been extensively studied in the context of one-dimensional (1D) wave propagation. Takezawa et al. (2000) assessed 1D-propagating regular wave interactions with a single trench

using the boundary element method, and showed that wave amplitude behind a trench can be reduced by 5–10%. McDougal et al. (1996) investigated 1D regular wave interactions with multiple pits or trenches and how they could be used to enhance wave attenuation, which they coined as “trench breakwaters”. Lee et al. (2009) further investigated wave attenuation by 1D propagating irregular waves due to interactions with both single and multiple trenches. These earlier studies identified wave reflection from trenches as a primary mechanism for wave attenuation. Additionally, multiple periodic trenches can generate Bragg reflection, further reducing wave amplitude behind the structures. Many researchers have designed various breakwater structures based on these studies (Barbosa-López et al., 2019; Kar et al., 2020; Gao et al., 2021, 2024; Xie, 2022; Hou and Gao, 2026). However, the aforementioned studies have largely focused on a one-dimensional analysis along the wave-propagation direction, thereby addressing wave incidence and reflection to/from laterally uniform trench geometries.

However, few studies have examined the interaction of two-dimensional (2D) propagating waves with trench-like bathymetry, which can alter wave transmission via refraction and/or diffraction. These mechanisms can also be harnessed to regulate wave-propagation directions, potentially allowing waves to bypass certain areas and

This article is part of a special issue entitled: Wave resonance phenomena published in Ocean Engineering.

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<https://doi.org/10.1016/j.oceaneng.2026.125903>

Received 23 December 2025; Received in revised form 26 February 2026; Accepted 2 May 2026

Available online 9 May 2026

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thereby reducing wave amplitude in those regions. These types of directional wave transmission devices are known as waveguides (Zhang et al., 2019). The concept of a waveguide was first introduced in the field of electromagnetic waves, where it refers to a device that can guide, propagate, and transmit electromagnetic waves with low energy loss (Bhandarkar, 2005; van der Tol et al., 2010). Due to the similarity between the shallow water wave equation and Maxwell's equations (Arthur et al., 1952), water waves can also be steered by waveguides. In recent years, researchers have utilized water wave 'metamaterials', defined as artificial (engineered) structures that alter the physical properties of the medium through which water waves propagate, to construct waveguides for the directional control of water waves to reduce wave amplitudes. Zou et al. (2019) designed a symmetric water wave waveguide attenuating structure by adjusting the water depth along the sides of a channel. In this structure, wave energy can be concentrated above the waveguide, while the wave height in the channel is reduced, resulting in the waveguide structure significantly decreasing wave heights within the channel. Other studies have explored the use of water wave metamaterials to create waveguide 'cloaks' that enable waves to bypass certain areas (Farhat et al., 2008; Wang et al., 2017; Zareei and Alam, 2015). However, water wave metamaterials require the application of spatial variation theory to create anisotropic water depths, which hardly exist in natural environments (Zhang et al., 2024). Consequently, the practical application of water wave metamaterial waveguides still faces challenges due to design uncertainty.

Research on wave manipulation using engineered seabed topography has evolved along two main strands: one focusing on wave guiding and trapping by ridges, and the other on wave refraction and attenuation by trenches. In studies concerning ridges, Buchwald (1969) initially proposed that bathymetric ridges could act as natural waveguides for long waves—a concept further advanced by Porter and Marangos (2022), who demonstrated that periodically arranged artificial ridges can directionally steer wave propagation. Wang et al. (2023) provided an analytical solution based on shallow water equations to explain the physical mechanism of wave trapping over ridges. On the other hand, several studies have highlighted the role of trenches in modulating wave fields through refraction. Kirby and Dalrymple (1983) employed potential flow theory to quantify changes in wave amplitude over rectangular trenches, establishing a theoretical foundation for subsequent research. Niu and Yu (2011) showed that circular trenches can induce significant wave amplitude reduction in their lee side, while Jung et al. (2008) and Liao et al. (2014) emphasized the importance of trench cross-section and sidewall slope on wave refraction patterns. Under more realistic 2D wave conditions, Williams and Vazquez (1991) identified pronounced shadow zones behind trenches, and Kim and Lee (2010) reported that wave incidence angle could alter attenuation efficiency by up to 20%, underscoring the need to account for wave direction in practical applications.

Collectively, these findings demonstrate that both ridges and trenches can effectively control wave propagation to achieve targeted wave reduction. To accurately evaluate the complex two-dimensional interactions between waves and submerged topographies, advanced numerical models are increasingly relied upon. Traditional depth-averaged shallow water equations and Boussinesq-type models, while computationally efficient, often face limitations in fully capturing the strong dispersive properties and complex vertical flow structures induced by abrupt bathymetric changes. Consequently, recent years have seen a rapid development and application of non-hydrostatic water wave models (e.g., NHWAVE, SWASH) in coastal engineering. Recent advancements in these models, particularly the implementation of multi-layer terrain-following coordinate systems and high-order shock-capturing schemes, have significantly improved their capability to resolve highly nonlinear wave transformation processes, including strong diffraction, wave breaking, and nonlinear energy transfer over complex coastal structures (Ai et al., 2014, 2019, 2021a, 2021c; Fang

et al., 2015; Qu et al., 2021; Ran et al., 2021).

Most importantly, these non-hydrostatic frameworks have demonstrated exceptional accuracy in simulating wave refraction and diffraction over variable seabed topographies. For instance, recent studies have extensively utilized non-hydrostatic models to investigate wave focusing over submerged shoals, dynamic interactions with trenches, and wave scattering around breakwaters, proving their reliability against classic experimental benchmarks (Zijlema et al., 2011; Ai et al., 2021b; David, 2022; Ma et al., 2012). Because the waveguide proposed in this study relies heavily on the precise control of wave propagation directions through secondary refraction and diffraction across depth discontinuities, adopting a multi-layer non-hydrostatic model is both justified and necessary to capture the full nonlinear hydrodynamics governing the sheltering performance.

However, previous research on the interaction between water waves and bathymetry changes has primarily focused on changes in waveforms after water waves pass through areas with varying water depths, without focusing on identifying how bathymetry changes can be optimized to act as a water waveguide.

To explicitly distinguish the contributions of this study from established methods, Table 1 summarizes the mechanisms and geometric characteristics of existing wave attenuation strategies. While traditional breakwaters provide robust protection, their seabed-mounted and surface-piercing nature imposes significant environmental and navigational constraints. Conversely, submerged trenches and ridges offer unobtrusive alternatives but primarily rely on reflection mechanisms that limit their effectiveness for directional steering. Although recent advances in metamaterial-based waveguides have demonstrated precise wave control, their reliance on complex, anisotropic bathymetry presents substantial challenges for practical implementation.

Synthesizing the limitations outlined in Table 1, there is a distinct lack of wave control solutions that combine the directional steering capabilities of metamaterials with the constructional simplicity of traditional submerged topography. To address this gap, this study designs a waveguide that combines traditional coastal engineering terrain features (such as trenches) with the law of refraction. This approach aims to address the complex structural challenges and practical engineering difficulties commonly encountered in existing water waveguides. Therefore, in the present study, a novel waveguide design that alters the propagation direction of waves based on the law of refraction is first proposed. The front of the waveguide is designed to form an angle with the incident wave, and the water depth inside the waveguide is set to exceed the external water depth, thereby modifying the refractive index of the water waves. These two aspects cause the waves to be refracted, thereby changing their propagation direction and bypassing the original propagation path; this, in turn, reduces the amplitude of the waves behind the waveguide.

This paper is organized as follows: In Section 2, the structure of the waveguide is designed according to the law of refraction. Section 3

Table 1
Comparison of mechanisms and limitations of existing wave control structures.

Structure Type	Primary Mechanism	Bathymetry Characteristics	Key Limitations
Traditional Breakwaters	Energy dissipation & Physical blocking	Seabed-mounted & Surface-piercing	High cost; navigational obstruction; environmental impact.
Submerged Trenches/Ridges	Bragg Reflection & Partial Trapping	Submerged; Simple/Periodic	Ineffective for 2D steering; relies on resonance conditions.
Metamaterial Waveguides	Transformation Optics (Anisotropy)	Complex, spatially varying (Anisotropic)	High design uncertainty; difficult to construct in natural seabed.

presents the numerical model, model configuration, and computational domain settings. Section 4 provides a comprehensive discussion of the numerical results of wave and waveguide interactions, including the refraction phenomenon that occurs after waves pass through wave guides, the wave sheltering regions formed behind wave guides with different inner and outer water depth ratios, and the changes in the sheltering regions formed after incident waves with different parameters pass through wave guides. Section 5 addresses the study's limitations and practical engineering considerations. Finally, Section 6 summarizes the primary conclusions and identifies potential avenues for future

investigation.

2. Waveguide design

The theoretical basis for the waveguide structure design is Snell's Law, given by:

$$\frac{\sin \alpha_0}{c_0} = \frac{\sin \alpha}{c} \quad (1)$$

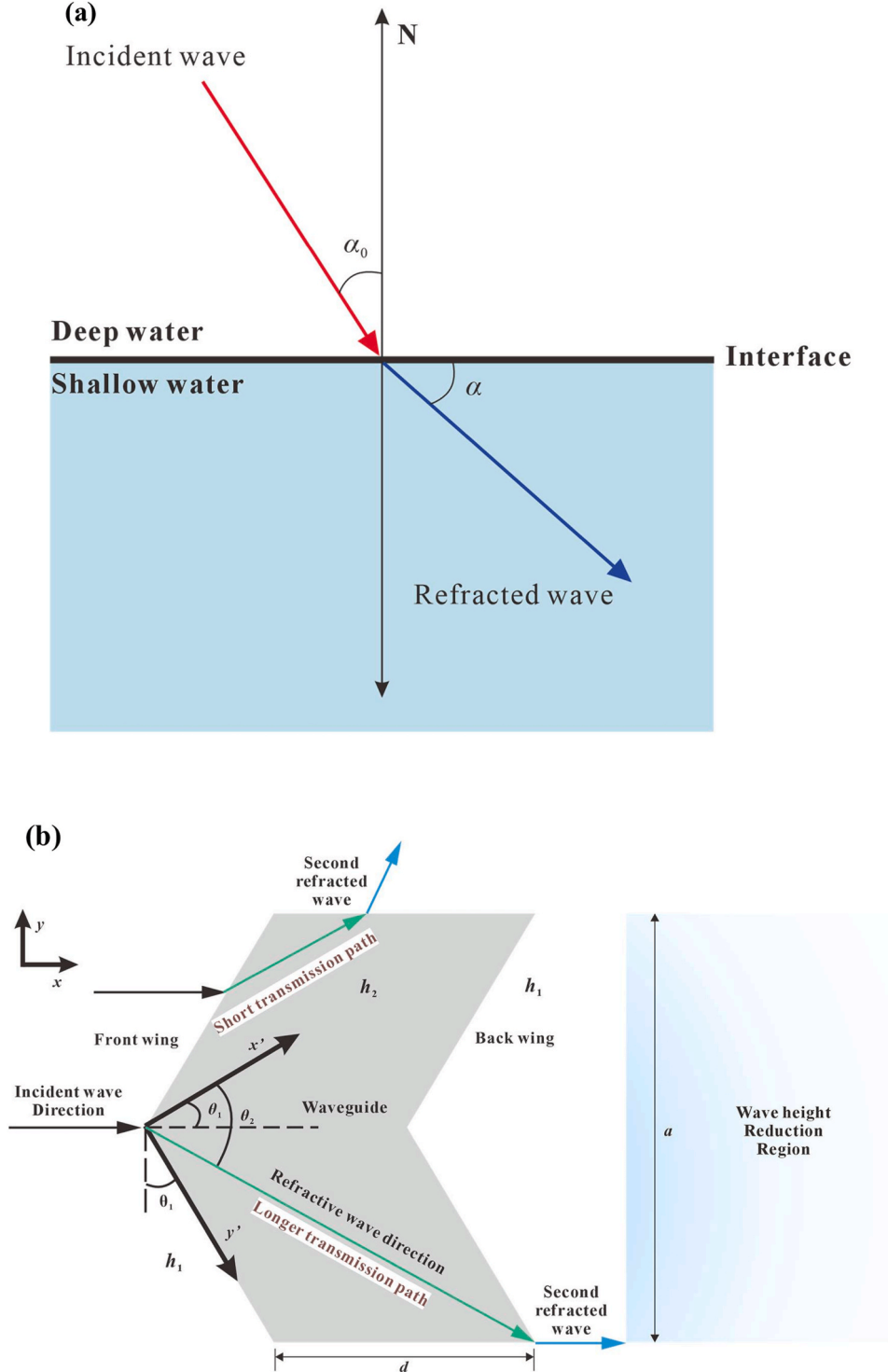


Fig. 1. (a) Schematic Diagram of Wave Refraction Process (b) Schematic diagram of Butterfly-shape waveguide.

where α_0 and c_0 are the angle of incidence and wave velocity in the incident medium, and α and c denote the angle of refraction and wave velocity in the refracting medium. For water waves, Eq. (1) is based on linear wave theory, where c and c_0 represent the phase velocity of the wave, determined by water depth and wave period. The angle of incidence and the angle of refraction refer to the angle between the direction of the incident wave and the direction of the refracted wave, and the normal vector of the interface between two propagation media.

To simplify the theoretical derivation and focus on the guiding mechanism, the following assumptions are made: (i) the fluid is inviscid and incompressible, and the wave motion is irrotational; (ii) the wave steepness is small enough that linear wave theory applies; (iii) the transition of water depth at the waveguide boundaries is assumed to be abrupt, and (iv) the effects of wave reflection and diffraction at the structure edges are neglected in this idealized analytical model.

According to the water-wave theory, when waves propagate from shallow water to deep water, their speed increases. As illustrated by the refraction process shown in Fig. 1, when the incident wave angle (the angle between the wave direction line and the normal vector of the contour line) is not zero, once the wave passes through the interface, the angle of the wave direction line in the other water depth region (i.e., the refraction angle) also changes. Therefore, to change the propagation direction of a wave train, two conditions must be met. The first is that there must be regions with different water depths. The other is that the angle between the wave propagation direction and the water depth contour line cannot be 90° .

Based on Eq. (1), for a region of crest width a , the wave can be made to bypass this area through the principles of refraction, resulting in a reduction in wave height within that region. It is posited that, by applying these principles, a waveguide can be designed through targeted modification of the water depth in the relevant area. In accordance with these aforementioned principles, a waveguide structure consisting of two symmetric parallelograms can be designed (Fig. 1b). Assuming a flat-bottomed bathymetry with a constant depth h_1 , and the depth of the waveguide is h_2 , where $h_2 > h_1$. As shown in Fig. 1b, the front side of the waveguide structure facing the incoming waves (propagating from left to right in the x -direction) is called the “front wing.” The direction along the front wing of the waveguide is defined as the y -axis, and the direction perpendicular to the front wing is defined as the x -axis. The angle between the direction of the front wing and the y -axis is denoted θ_1 , and according to the geometrical conditions, the angle between the wave direction line of the incident wave and the x -axis is also equivalent to θ_1 . Due to the change in water depths inside and outside the waveguide, the waves become refracted as they pass through the waveguide front wing. When waves propagate to the upper or lower boundaries of the waveguide, they return from the deeper zone to the shallow zone. This causes a secondary refraction as the waves leave the waveguide boundary. According to the law of refraction, this portion of the wave does not propagate in the x -direction, thereby deflecting waves away from a specified area. However, when waves are incident at the midpoint of the waveguide's front wing, they propagate along a longer path due to refraction. If they cannot propagate to the upper or lower boundaries of the waveguide, they will propagate to the rear boundary, which is defined as the ‘back wing’. Since the back wing is parallel to the front wing, the waves will refract to resume their original propagation direction along the x -axis and enter the designated area. To avoid this, the length d of the waveguide should be extended along the x -axis until the longest refracted wave also exits the upper and lower edges of the waveguide. Therefore, the dimensions of the waveguide depend on the characteristics of the incident wave, as well as the incident and refraction angles. The following derivation uses wave refraction theory to determine the optimum dimensions of the waveguide structure.

According to the geometry of the waveguide in Fig. 1b, the angle of incidence θ_1 of the wave relative to the front wing of the waveguide corresponds to α_0 Eq. (1). Therefore, the following relationship should hold:

$$\frac{\sin \theta_1}{c_1} = \frac{\sin \theta_2}{c_2} \quad (2)$$

where θ_2 is the angle of refraction (i.e., the angle between the refracted wave direction line and the x -axis), and c_1 and c_2 correspond to the phase velocities propagating at water depths h_1 and h_2 , respectively. Divide both sides of the equation by the wave period T , Eq. (2) becomes:

$$\frac{\sin \theta_1}{c_1 T} = \frac{\sin \theta_2}{c_2 T} \quad (3)$$

in terms of the wave number $k = 2\pi/L$, where $L = cT$ is the wavelength. Based on Eqs. (2) and (3), the refraction at the interfaces of the waveguide satisfies:

$$k_1 \sin \theta_1 = k_2 \sin \theta_2 \quad (4)$$

where k_1 and k_2 are the wave numbers in the respective water depths h_1 and h_2 , with wave numbers governed by the dispersion relation:

$$\omega^2 = gk \tanh kh \quad (5)$$

As shown in Fig. 1, the waveguide structure has a width a in the y -direction. According to the above derivation and geometrical relationship, the length of the waveguide structure in the propagation direction, d , should satisfy the following relationship:

$$d = \frac{a}{2} \left(\frac{1}{\tan(\theta_2 - \theta_1)} - \tan \theta_1 \right) = \frac{a}{2} \left(\frac{1 + \frac{\sin \theta_2}{\sqrt{1 - \sin^2 \theta_2}} \tan \theta_1}{\frac{\sin \theta_2}{\sqrt{1 - \sin^2 \theta_2}} - \tan \theta_1} - \tan \theta_1 \right) \\ = \frac{a}{2} \left(\frac{1 + \frac{k_1 \sin \theta_1}{k_2} \tan \theta_1}{\sqrt{1 - \left(\frac{k_1 \sin \theta_1}{k_2} \right)^2} - \tan \theta_1} - \tan \theta_1 \right) \quad (6)$$

Therefore, based on Eq. (6), it is possible to determine the dimensions of the waveguide structure to provide wave sheltering. Fig. 2 shows the relationship between the dimensionless waveguide length d/a and the wave number ratio k_1/k_2 for different incident angles at front-wing θ_1 . According to the dispersion relation in Equation (5), when the incident wave period is constant, the water depth h exhibits an

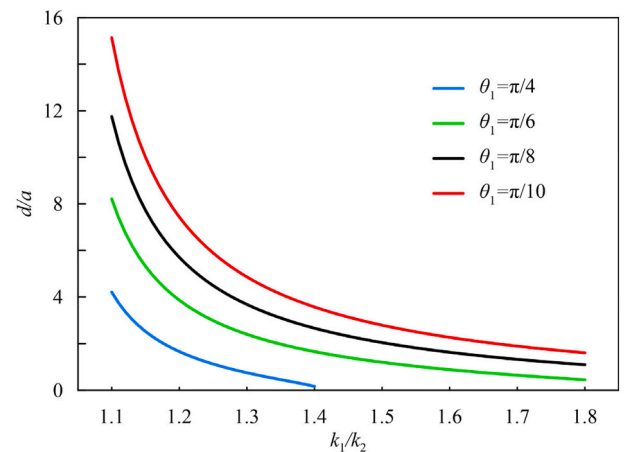


Fig. 2. Relationship between the dimensionless waveguide length d/a and the wave number ratio k_1/k_2 for different incident angles at front-wing θ_1 .

inverse relationship with the wave number k . Therefore, a larger ratio of k_1/k_2 indicates a greater difference in water depth between the waveguide interior and exterior. It can thus be observed that, for a fixed incident angle at the front-wing θ_1 , the greater the water depth ratio, the shorter the required design length of the waveguide. In addition, the results show that for a fixed water depth ratio, the greater the incident angle at the front-wing, the shorter the required design length.

The relationship given in Eq. (6) is exclusively based on properties of the wave interactions with the waveguide governed by refraction predicted by linear wave theory. However, when waves pass through varying terrain, they can undergo additional phenomena, including reflection and diffraction. To investigate the effect of waveguides on wave sheltering in the lee area under actual conditions, this theoretical framework is compared with results obtained from a numerical non-hydrostatic wave model that simulates the complete nonlinear interactions between waveguides and water waves.

It should be noted that while this analytical model provides a deterministic design tool based purely on refraction, the actual performance in a real sea state may be influenced by nonlinear interactions. Specifically, when waves encounter the depth-discontinuity of the waveguide, secondary phenomena such as diffraction and partial reflection may occur. Consequently, to validate the efficacy of the design under more complex conditions, this theoretical framework is subsequently compared with a numerical non-hydrostatic wave model that accounts for the complete nonlinear physics.

3. Numerical simulation

3.1. Non-hydrostatic water wave model

In this study, simulations are carried out using the modified non-hydrostatic water wave model (He et al., 2020). Referring to Ma et al. (2012), the Euler equations in Cartesian coordinates (x, y, z) are transformed to σ -coordinates (x, y, σ) . With this transformation, the z -coordinate is transformed into σ -coordinate space where σ spans 0 to 1 with $\sigma = 0$ coinciding with the bottom and $\sigma = 1$ with the surface, i.e.:

$$\sigma = \frac{z + h}{D} \quad (7)$$

where the total water depth $D(x, y, t) = h(x, y) + \eta(x, y, t)$. In this coordinate system, the governing Euler equations are:

$$\frac{\partial D}{\partial t} + \frac{\partial Du}{\partial x} + \frac{\partial Dv}{\partial y} + \frac{\partial \tilde{w}}{\partial \sigma} = 0 \quad (8)$$

$$\frac{\partial \mathbf{U}}{\partial t} + \frac{\partial \mathbf{F}}{\partial x} + \frac{\partial \mathbf{G}}{\partial y} + \frac{\partial \mathbf{H}}{\partial \sigma} = \mathbf{S}_h + \mathbf{S}_p \quad (9)$$

where $\mathbf{U} = (Du, Dv, Dw)^T$; u and v are the horizontal flow velocities in the x , y , respectively; $\tilde{w}$ is the vertical flow velocity in the σ coordinate system,

$$\tilde{w} = D \left(\frac{\partial \sigma}{\partial t} + u \frac{\partial \sigma}{\partial x} + v \frac{\partial \sigma}{\partial y} + w \frac{\partial \sigma}{\partial z} \right) \quad (10)$$

in which w denotes the velocity component in the z -direction, and the

$$\begin{aligned} \frac{\partial \sigma}{\partial t} &= \frac{\sigma}{D} \frac{\partial D}{\partial t} \\ \frac{\partial \sigma}{\partial x} &= \frac{\partial h}{\partial x} - \frac{\sigma}{D} \frac{\partial D}{\partial x} \\ \frac{\partial \sigma}{\partial y} &= \frac{\partial h}{\partial y} - \frac{\sigma}{D} \frac{\partial D}{\partial y} \\ \frac{\partial \sigma}{\partial z} &= \frac{1}{D} \end{aligned} \quad (11)$$

$\mathbf{F}$, $\mathbf{G}$, $\mathbf{H}$ are the fluxes,

$$\mathbf{F} = \begin{pmatrix} Duu + \frac{1}{2}gD^2 \\ Duu \\ Duw \end{pmatrix} \quad \mathbf{G} = \begin{pmatrix} Duv \\ Dvv + \frac{1}{2}gD^2 \\ Dvw \end{pmatrix} \quad \mathbf{H} = \begin{pmatrix} u\tilde{w} \\ v\tilde{w} \\ w\tilde{w} \end{pmatrix}$$

and S_h , S_p represent the bottom-slope source term, and the non-hydrostatic pressure term, respectively.

$$\mathbf{S}_h = \begin{pmatrix} gD \frac{\partial h}{\partial x} \\ gD \frac{\partial h}{\partial y} \\ 0 \end{pmatrix} \quad \mathbf{S}_p = \begin{pmatrix} -\frac{D}{\rho} \left(\frac{\partial p}{\partial x} + \frac{\partial p}{\partial \sigma} \frac{\partial \sigma}{\partial x} \right) \\ -\frac{D}{\rho} \left(\frac{\partial p}{\partial y} + \frac{\partial p}{\partial \sigma} \frac{\partial \sigma}{\partial y} \right) \\ -\frac{1}{\rho} \frac{\partial p}{\partial \sigma} \end{pmatrix}$$

where the total pressure has been divided into two parts: dynamic pressure p (use p as dynamic pressure hereinafter for simplicity) and hydrostatic pressure $\rho g(\eta - z)$.

Integrating Eq. (8) from $\sigma = 0$ to 1, and using the boundary conditions at the bottom and surface for $\tilde{w}$, we obtain the governing equation relating water depth and flow velocities:

$$\frac{\partial D}{\partial t} + \frac{\partial}{\partial x} \left(D \int_0^1 u d\sigma \right) + \frac{\partial}{\partial y} \left(D \int_0^1 v d\sigma \right) = 0 \quad (12)$$

Eqns. (8)–(12) are solved using a mixed finite volume and finite difference discretization scheme. The flux term of the equation is discretized using the finite volume method, while the source term on the right-hand side of the equation is discretized using a center difference scheme with second-order accuracy. To accurately apply pressure boundary conditions on the free surface, the model employs staggered grids for discretization, where velocity is defined at the center of the grid and pressure is defined on the surface of the grid.

The model applies a high-order reconstruction method (a cubic weighted essentially non-oscillatory, or WENO scheme (Kurganov and Levy, 2000)) to accurately represent wave variables in the horizontal direction without introducing artificial oscillations. Fluxes across grid cell boundaries are computed using the MUSTA approximate Riemann solver (Toro and Titarev, 2006), which efficiently captures nonlinear wave propagation. Time integration is carried out using a second-order strong-stability Runge-Kutta scheme (Derakhti et al., 2015), ensuring both accuracy and stability when advancing the solution in time.

In the simulations, adaptive time steps Δt are adopted that satisfy the Courant Friedrichs Lewy (CFL) criterion, in which Δt varies at each time step to meet:

$$\Delta t = C \min \left[\min \frac{\Delta x}{|u_{i,j,k}| + \sqrt{gD_{i,j}}}, \min \frac{\Delta y}{|v_{i,j,k}| + \sqrt{gD_{i,j}}}, \min \frac{\Delta \sigma D_{i,j}}{|w_{i,j,k}|} \right] \quad (13)$$

At the model boundaries, a numerical sponge layer based on Larsen & Darcy (Larsen and Dancy, 1983) is included to minimize wave reflections. The sponge layer is imposed over a range $0 < x < x_s$ within the range, the damping coefficient γ has the following relationship:

$$\gamma(x) = \begin{cases} \exp \left[\left(\frac{2 - x/x_s}{2 - x_s/x_s} \right) \ln a \right], & 0 < x < x_s \\ 1, & x \geq x_s \end{cases} \quad (14)$$

where x_s is the width of the sponge layer, defined as the distance from the starting position of the sponge layer to the open boundary, and here a is a constant that depends on the number of mesh points in the sponge layer ($x_s/\Delta x$), and is used to control the rate of attenuation.

3.2. Model setup and parameter space

The simulation in this study is a three-dimensional simulation. The

plan view of the computational domain is shown in Fig. 3. The waveguide structure was positioned approximately 7 wavelengths downstream of the wave-generation boundary and extended approximately 9 wavelengths downstream to observe wave evolution. To avoid interference from residual reflections on both sides of the lateral boundaries, the distance from the waveguide to the lateral boundary of the computational domain was 10 wavelengths. Numerical sponge layers were installed on the lateral and downstream boundaries, which extended over a distance approximately 3 times the wavelength.

The water depth outside the waveguide structure is h_1 , and the water depth inside the waveguide structure is h_2 . Different water depth ratios were used to investigate the influence of changes in water depth inside and outside the waveguide structure on the waveguide effect. In the simulations, a default waveguide geometry was used to investigate general features of wave interactions, with parameters subsequently varied to assess responses to varying properties. For the simulations, the default waveguide geometry was defined with a front-wing angle of $\pi/6$, with the water depth outside the waveguide $h_1 = 0.4$ m. The total width of the waveguide structure in the y-direction is denoted by a , which, for the default geometry, was set to be equal to the incident wavelength 3.7 m. Parameters of waveguides in simulation are shown in Table 2. According to the CFL criterion in Eq. (13), C is set to 0.5 in the simulation.

3.3. Model validation and grid convergence

To validate the accuracy of the non-hydrostatic model in simulating wave refraction and diffraction over complex bathymetry—mechanisms that are fundamental to the proposed waveguide design—the numerical model is evaluated against the classic elliptical shoal experiment conducted by Berkhoff et al. (1982). The experimental setup features an elliptical shoal situated on a plane beach with a slope of 1/50. The boundary of the shoal is mathematically defined by $(x'/3)^2 + (y'/4)^2 = 1$. The schematic diagram of the geometric structure at the bottom of the

Table 2

Parameter space.

Type	Variable	Notation	Values (default)	Units
Waveguide Properties	Waveguide length	d	3.88-38.53 (16.00)	m
	Incident angle at front-wing	θ_1	$\pi/10 - \pi/4$ ($\pi/6$)	-
	Waveguide width	a	3.7	m
	Water depth outside the waveguide	h_1	0.4	m
	Water depth inside waveguide	h_2	0.48-1.4 (0.6)	m
	Ratio of water depth inside & outside waveguide	h_2/h_1	1.2-3.5 (1.5)	-
Wave Parameters	Incident wave height	H_0	0.02-0.12 (0.02)	m
	Incident wave period	T	1.4-3.6 (2.0)	s

Note that values in the parentheses denote the default/reference values described in the text.

experiment is shown in Fig. 4. Regular waves with a period of 1.0 s and an incident wave height of 4.64 cm are generated at the boundary.

For the computational configuration, a fine grid with spatial resolutions of 0.025 m and 0.05 m in the x and y directions, respectively, is utilized. Crucially, the vertical domain is discretized into three vertical layers.

Comparison of simulated wave heights at cross-sections (1) to (8) with experimental results is shown in Fig. 5. The simulation results demonstrate that the numerical model successfully captures the wave focusing phenomenon behind the shoal, predicting a maximum wave height approximately 2.2 times the incident wave height. The simulated spatial wave height variations across multiple measurement sections exhibit excellent agreement with the experimental data. This validation

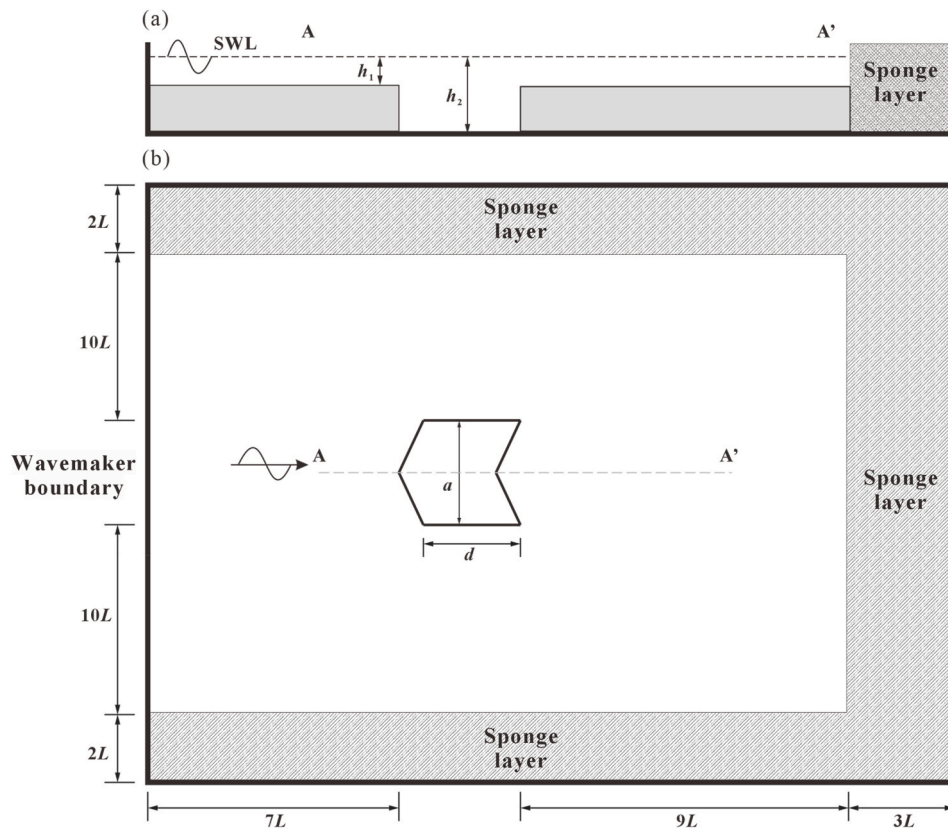


Fig. 3. Schematic diagram of calculation domain (a) A-A' sectional side view (b) top view of calculation domain.

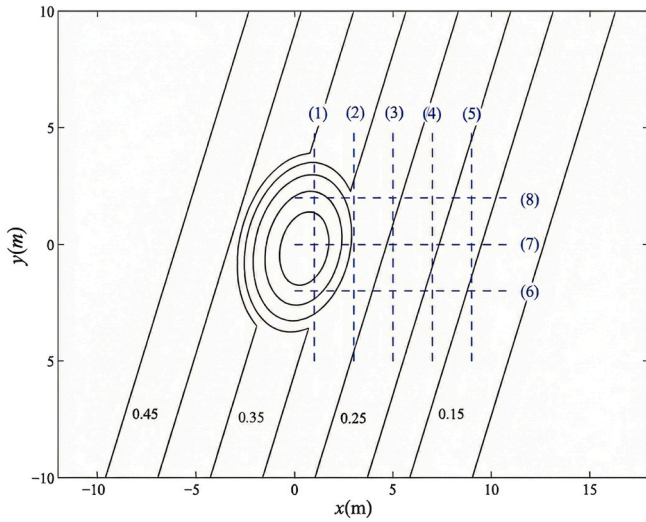


Fig. 4. Bottom geometry for periodic wave propagation over an elliptical shoal, experimental setup by Berkhoff et al. (1982).

confirms that the model, specifically when utilizing three vertical layers, possesses the requisite numerical accuracy to reliably simulate the complex wave transformation processes relevant to the waveguide structure.

Given the large propagation area and long duration of wave action, ensuring the numerical model's stability in wave generation and grid convergence are crucial. Therefore, prior to numerical simulations, it is essential to perform grid convergence testing on the selected model to ensure the accuracy and reliability of the computational results.

The non-hydrostatic model employed three vertical layers of grids to adequately simulate wave variation (Ma et al., 2012). In the horizontal

direction, to assess the impact of grid size on wave-propagation accuracy, five grid resolutions were initially evaluated, corresponding to 1/60, 1/80, 1/100, 1/120, and 1/140 of the wavelength. Convergence analysis was performed using different grid sizes to verify numerical stability and accuracy during wave propagation.

The convergence analysis was conducted for the wave condition with an incident wave period of $T = 2.0$ s and wave height $H = 0.02$ m, comparing the temporal wave surface curves within the computational domain across the five aforementioned mesh sizes. To quantify the variability in computational results across different grid parameters, based on the calculation of Willmott model skill (Willmott, 1981):

$$Skill = 1 - \frac{\sum_{i=1}^N |\eta_{i,a} - \eta_{i,b}|^2}{\sum_{i=1}^N (|\eta_{i,a} - \bar{\eta}_b| + |\eta_{i,b} - \bar{\eta}_b|)^2} \quad (15)$$

where a and b denote wave surface data for different grid parameters; $\eta_{i,a}$ and $\eta_{i,b}$ denote the wave surface elevation at time step i for grid parameters a and b , respectively; $\bar{\eta}_b$ denotes the time-series mean of the wave surface for grid parameter b . The value of $Skill$ ranges from 0 to 1. $Skill = 1$ indicates that the two wave surface data sets are entirely consistent, while $Skill = 0$ means a significant discrepancy.

Fig. 6 shows how the $Skill$ of the predicted water surface elevation time series along the A-A' section in Fig. 3 varies for the simulations at five different grid sizes. For ease of comparison, the b values in the $Skill$ function were uniformly set to wave surface data from a 1/140 wavelength grid. It can be observed that the wavefronts exhibit little differences across different grid sizes, both in front of the waveguide and in the mid-front region. However, behind the waveguide, wavefront differences between grid sizes begin to emerge due to refraction. Notably, results for the 1/60- and 1/80-wavelength grid sizes differ from those obtained with denser grid sizes. When the grid density increases beyond 1/100 wavelength, the $Skill$ value alongside the waveguide approaches

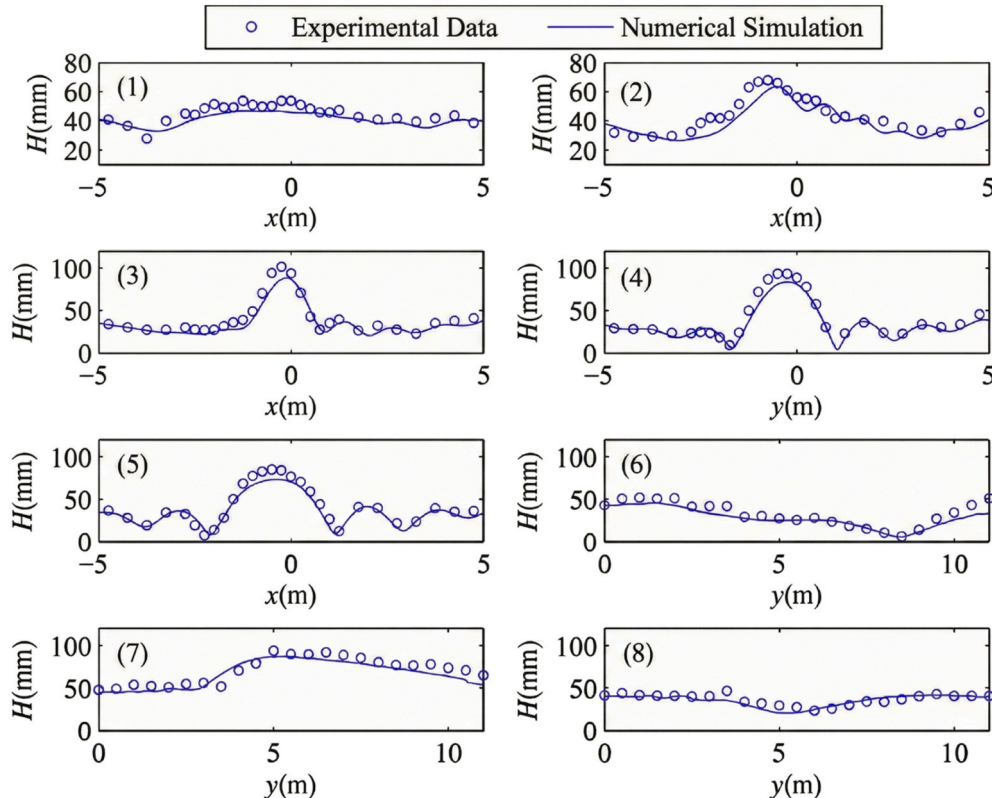


Fig. 5. Comparisons between numerical and experimental wave height at eight sections.

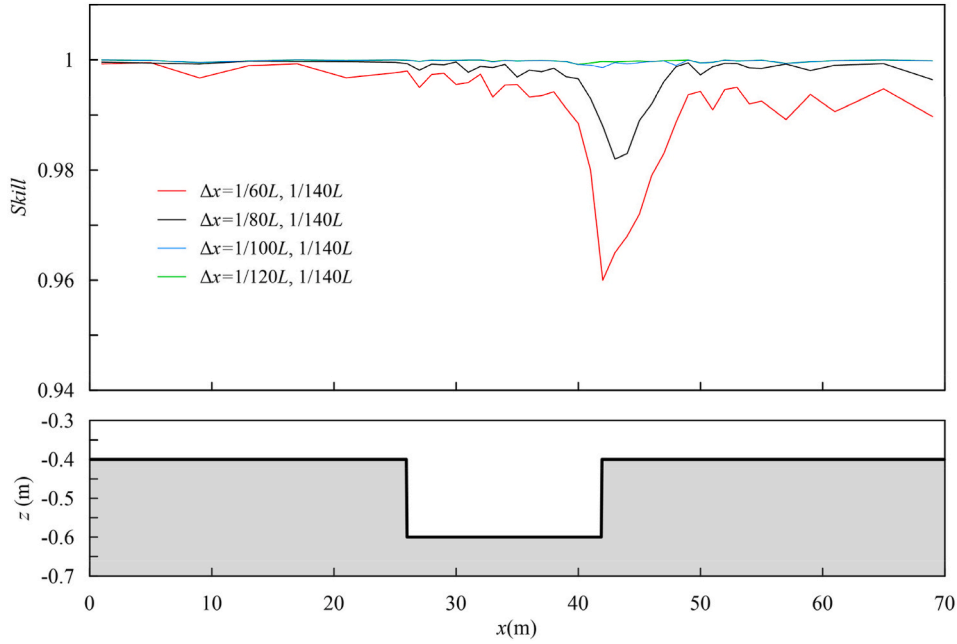


Fig. 6. Distribution of the wave surface similarity coefficient *Skill* along the waveguide cross-section A-A' in Fig. 2 for different grid sizes (1/60 to 1/140 wavelength).

1. This indicates that beyond a grid density of 1/100 wavelength, wave surface differences become negligible, and further increasing the grid resolution does not affect computational results.

Based on the convergence analysis results, the grid size of 1/100 of the wavelength ensures both computational accuracy and effective control of computational cost. Therefore, the 1/100 wavelength grid size was selected as the final configuration for the simulations in this study, meeting the precise computational requirements for wave propagation and ensuring the stability of the numerical results.

4. Results and discussion

4.1. Wave propagation through the waveguide

An example case, based on the default wave condition with $H_0 = 0.02$ m, $T = 2.0$ s in Tables 1 and is first used to illustrate the sheltering

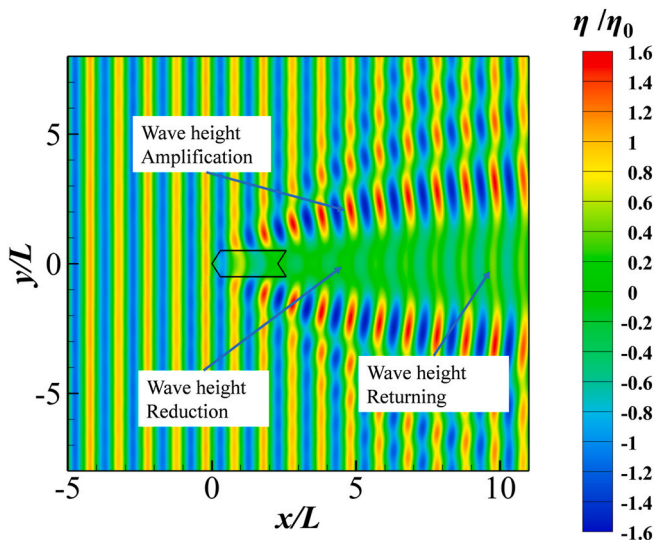


Fig. 7. Changes in waves after passing through a waveguide, visualized by a snapshot of the water surface elevation.

effect downstream of the waveguide at different depth ratios (h_2/h_1). Fig. 7 shows the distribution of the stabilized wavefield at a depth ratio of $h_2/h_1 = 2.5$, and it can be seen that when the incident wave passes through the waveguide, significant wave refraction occurs, leading to a notable reduction in the downstream wave height. In addition, near the edge of the shelter region, wave height amplification forms a fan-shaped pattern. The simulation results demonstrate that wave height attenuation in the lee is consistent with the waveguide's design objectives, as outlined in Section 2. However, the increase in waves at the rear of the waveguide is caused by the interaction between the waves refracted from the waveguide side boundary and the waves outside the waveguide. We next focus on quantifying how the incident and refracted wave patterns control the geometry of the sheltered and amplification region.

As illustrated in Fig. 8, the propagation paths of waves incident on the front wing of a waveguide are refracted towards the lateral boundary of the waveguide. In the context of wave propagation from deep to shallow water within a waveguide, it is postulated that the incident angle of the refracted wave η_2 at the lateral boundary is θ_3 , and that the refraction angle of the wave η_3 exiting the lateral boundary is θ_4 . In accordance with the law of refraction (Eq. (4)), the wave angles across

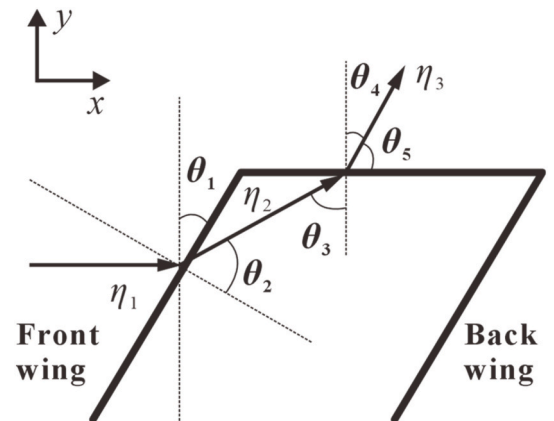


Fig. 8. Schematic diagram of the incident and refracted wave paths through the waveguide.

the lateral boundary are:

$$k_2 \sin \theta_3 = k_1 \sin \theta_4 \quad (16)$$

k_1 is the wave number outside the waveguide (determined by h_1), and k_2 is the wave number inside the waveguide (determined by h_2). According to the geometric relationship, the above angles are related as follows:

$$\theta_3 = \frac{\pi}{2} - (\theta_2 - \theta_1) \quad (17)$$

Therefore, the refraction angle of the trailing wave η_3 is θ_4 :

$$\theta_4 = \arcsin\left(\frac{k_2 \cos(\theta_2 - \theta)}{k_1}\right) \quad (18)$$

According to the dispersion relation, the water depth h is inversely proportional to the wave number k . When the water depth in the waveguide increases as the water depth outside the waveguide remains constant, the wave number ratio between inside and outside waveguide k_2/k_1 decreases, leading to a decrease in θ_4 . However, θ_4 is the angle between the wave direction line and the normal to the lateral boundary (y -axis). Based on the above geometric relationships, the theoretical emission angle (θ_5) of the waves emitted from the side, also equivalent to the difference in angle between the incident wave direction, is:

$$\begin{aligned} \theta_5 &= \frac{\pi}{2} - \theta_4 \\ \theta_5 &= \frac{\pi}{2} - \arcsin\left(\frac{k_2 \cos(\theta_2 - \theta)}{k_1}\right) \end{aligned} \quad (19)$$

Thus, when the angle of incidence θ remains constant, the ratio of water depth between inside and outside the waveguide increases, θ_4 decreases and θ_5 increases. The closer the direction of waves emitted from the lateral boundary is to the y -axis (orthogonal to the incident wave direction), the larger the sector in which wave height increases.

Fig. 9 shows the normalized wave height distribution after passing through the waveguide designed with the default incident angle at the front-wing of $\pi/6$, and the ratio of water depth inside and outside the waveguide (h_2/h_1) ranging from 1.5 to 3.5. According to the designed waveguide length b derivation in Eq. (6) and Fig. 2, as the designed water depth inside the waveguide increases, the waveguide length decreases accordingly. As shown, the wave amplitude decreases behind the waveguide, indicating that the waveguide effectively attenuates waves. As the water depth of the waveguide increases, the angle of the sector in which the wave height increases become larger, and the range of influence in the y -direction expands too. This is consistent with the aforementioned derivation of the direction of secondary-refraction waves in the waveguide. Next, the relationship between the sector angle and the theoretically derived values will be analyzed.

As shown in Fig. 10, a fan-shaped region can be observed on both the upper and lower sides of the waveguide, where the wave height begins to vary compared to the stable propagation region ahead of the waveguide. The boundary of this fan-shaped area is highlighted using the two dashed lines. The fan-shaped regions will be referred to as wave-affected regions. This region is formed by the secondary refraction of waves at the side boundary of the waveguide and the convergence of waves in the x -direction. The angle between the upper edge of the fan-shaped region and the x -direction will be denoted as β . Based on the derivation of the wave propagation direction Eq. (19), β should be approximately equal to θ_5 , since this represents the maximum angle at which refracted waves can propagate.

Fig. 11 illustrates the relationship between β and θ_5 , as well as the water depths inside and outside the waveguide. As shown, consistent with the theoretical derivation, both β and θ_5 increase as the ratio of the water depths inside and outside the waveguide increases. However, β is always greater than θ_5 , indicating that waves emitted from the lateral boundary interact with the directly incident waves, thereby influencing the wave field over a wider area, which causes the angle β of the sector to

be greater than θ_5 . In addition, as h_2/h_1 increases, the difference between β and θ_5 also increases. This indicates that the influence of waves emitted from the lateral boundary on the wave field is positively correlated with θ_5 . As the water depth increases, the range of influence of waves emitted from the lateral boundary of the waveguide on the wave field in the y -direction increases too.

To quantify sheltering effectiveness, the relative wave height H/H_0 is introduced, where H and H_0 represent wave heights downstream and upstream of the waveguide, respectively. As shown in Fig. 9, the wave heights are substantially attenuated in all cases, with the maximum attenuation exceeding 90%, which corresponds to the region where $H/H_0 < 0.1$ in the diagram. These results confirm the waveguide's potential as a high-efficiency wave protection system. However, according to Fig. 9, the sheltering effect exhibits spatial variations, gradually diminishing along the direction of wave propagation further in the lee of the waveguide. This mechanism may be due to stronger wave diffraction at the back wing as the water depth within the waveguide increases, demonstrating the complex interplay between bathymetry and wave scattering.

Following McDougal et al. (1996), which defines breakwaters as being effective if wave heights in the sheltered zone are attenuated to below 40% of incident levels ($H/H_0 < 0.4$), we adopted a similar definition here. A sector-shaped ring-like area is formed by the sheltered zone, as evidenced by the simulation results. As illustrated in the Fig. 12, the length of the area where $H/H_0 < 0.4$ is designated as b , and the maximum width of the outer side of the sector-shaped ring is equivalent to the width of the sheltered zone $a_{\max}$.

Fig. 13 illustrates the relationship between normalized sheltered zone length (b/L_1) and waveguide length (d/L_1) with respect to varying depth ratios (h_2/h_1) at a fixed incident period ($T = 2.0$ s) and front-wing angle ($\theta = \pi/6$). Where L_1 denotes the wavelength of this incident wave in water depth h_1 . Increasing h_2/h_1 reduces both d/L_1 and b/L_1 , with the rate of reduction in b/L_1 diminishing significantly beyond $h_2/h_1 = 1.7$. However, the normalized width of the sheltered zone, $a_{\max}/L_1$, does not vary monotonically, initially increasing and subsequently decreasing, reaching a maximum of 2.12 when $h_2/h_1 = 2.2$, and then decreasing as the water depth ratio increases. These findings suggest that when the water depth ratio exceeds 2.2, the waveguide's effectiveness in sheltering gradually diminishes.

As shown in Section 2, the angle of incidence is a key factor in determining the refraction of waves. For the waveguide designed in this study, the angle between the front wing and the wave represents the angle of incidence. Consequently, the front-wing angle (θ_1) emerges as a critical yet secondary design parameter. Fig. 14 demonstrates that at fixed $h_2/h_1 = 1.5$, when θ_1 increases from $\pi/10$ to $\pi/4$, d/L_1 decreases roughly linearly by 58%, b/L_1 decreases by 43%, and the normalized sheltered zone $a_{\max}/L_1$ decreases by 66%. This strong angular dependence underscores the necessity of multi-parameter optimization, which is the focus of the following sections.

4.2. Analysis of a fixed-size waveguide for the influence of the incident wave

In the preceding sections, an analysis was conducted of the effects of waveguides designed according to the law of refraction on waves incident at the same wave period on wave refraction and lee sheltering. However, in actual engineering applications exposed to variable sea states, wave periods are naturally variable. It is thus essential to conduct a comprehensive analysis of the sheltering capabilities of waveguide trenches for incident waves of varying periods. In this scenario, a fixed waveguide size was utilized, with waves of varying periods being incident to investigate the sheltering effects of the same waveguide. In this section, simulated wave periods range from 1.4 s to 3.6 s, with the simulated wave height held constant at 0.04 m.

The fixed waveguide dimensions employ the default waveguide design parameters in Table 2. This finding suggests that the waveguide

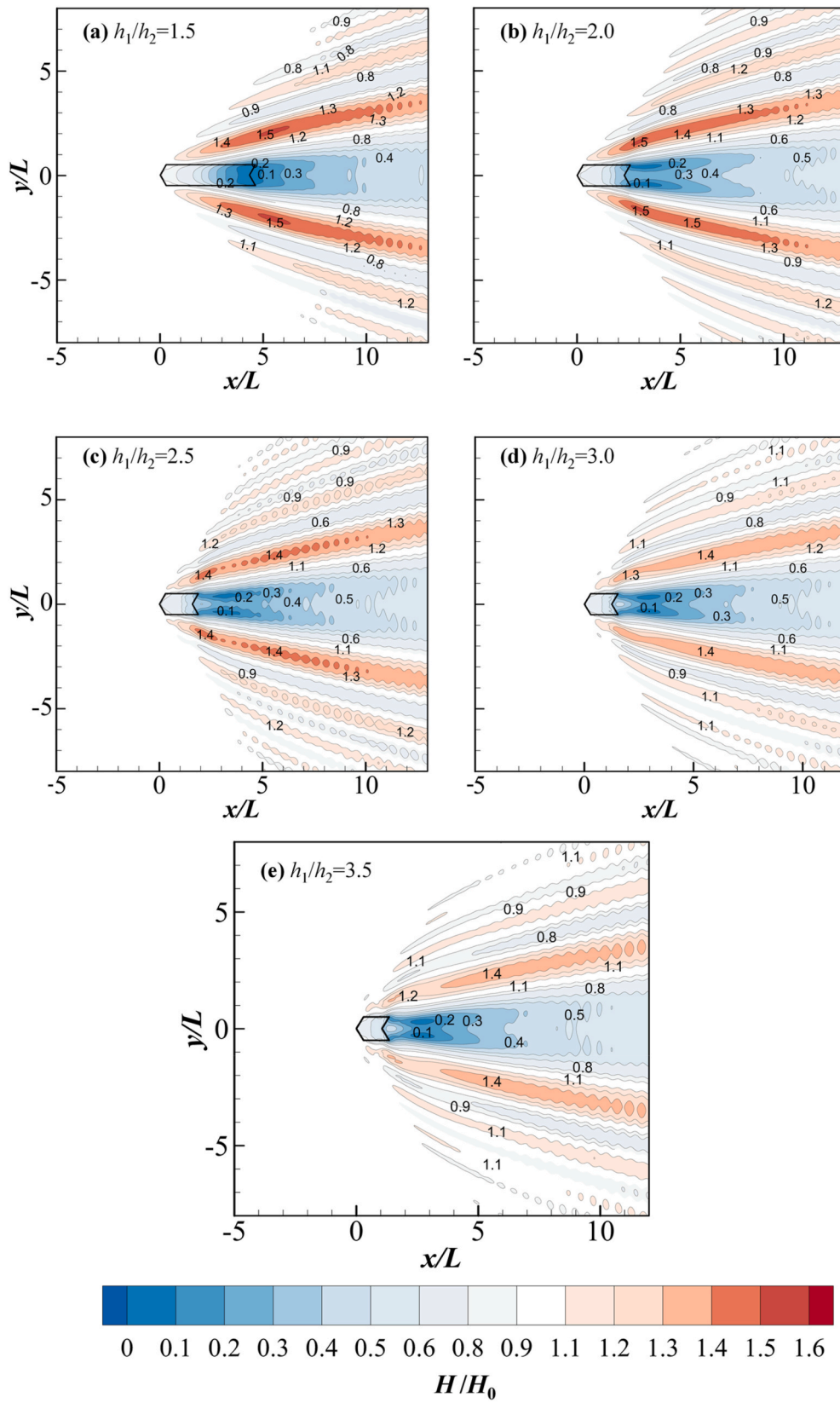


Fig. 9. Normalized wave height distribution around the waveguide (solid line), the front wing angle is fixed at 30° . Incident wave period $T = 2.0$ s (a) $h_2/h_1 = 1.5$, (b) $h_2/h_1 = 2.0$, (c) $h_2/h_1 = 2.5$, (d) $h_2/h_1 = 3.0$, (e) $h_2/h_1 = 3.5$.

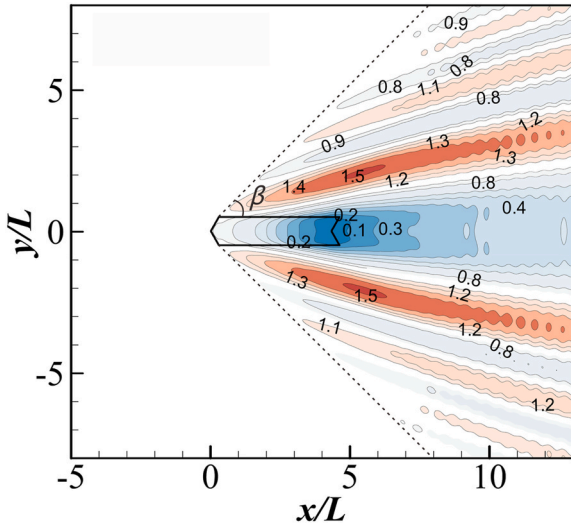


Fig. 10. Schematic diagram of the wave region near the waveguide.

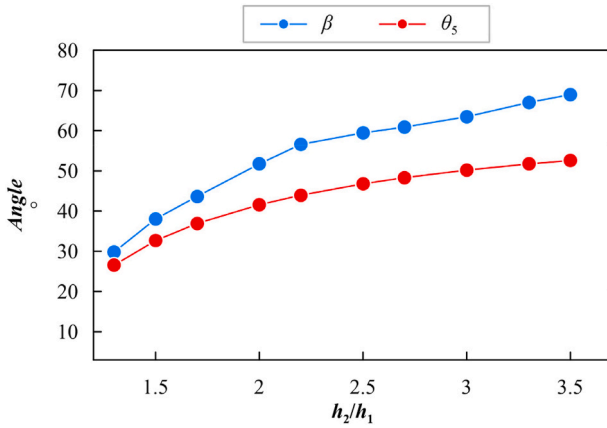


Fig. 11. The relationship between β and θ_s and the ratio of water depth inside and outside the waveguide h_2/h_1 .

trench exhibits wave sheltering effects at the lee of the waveguide under different wave period conditions.

According to the definition provided in the preceding section, the area of the sheltered zone at the lee of the waveguide is equivalent to the area where the relative wave height is $H/H_0 < 0.4$. Due to variations in the incident wave period, the wavelength of the incident wave L_1 also changes accordingly. However, in this section, the waveguide dimensions remain constant. To clearly illustrate how the dimensions of the sheltered zone vary with the incident wave period, in this section, parameters of the waveguide employed are used to render the dimensions of the shielded area dimensionless: dimensionless sheltered

zone length is denoted as b/d and the dimensionless maximum width of the sheltered zone as $a_{\max}/a$. The lengths of the non-dimensional sheltered zone formed by the incident waves of different periods are shown in Fig. 15. It is evident from these results that the length of the sheltered area behind the waveguide trench demonstrates a strong decreasing trend with the increase of the wave period, and the length of the sheltered area of the incident wave with incident wave period T . It is evident from the data that the $T = 3.6$ s configuration exhibits a 90% reduction in length compared with the sheltered area of the incident wave with $T = 1.4$ s. Similarly, as the cycle increases, the normalized width of the cover area $a_{\max}/L_1$ also decreases, with $T = 3.6$ s decreasing by 87% compared to $T = 1.4$ s. The occurrence of this phenomenon is closely related to the refraction angle. According to the derivation in Section 2, when the ratio of the inner and outer water depths and the incident angle are constant, the longer the wavelength of the incident wave, the smaller the refraction angle. Therefore, for a waveguide of a single size, the longer the period of the incident wave, the longer its wavelength, meaning that more waves will not exit from the upper and lower edges of the waveguide. Instead, they will exit from the back-wing position of the waveguide. In this situation, the waves emitted from the back-wing combine with diffracted waves, which reduces the shelter effect at the rear.

Fig. 16 shows how the dimensions of the sheltered area behind the waveguide vary according to different wave heights at a constant incident wave period of $T = 2.0$ s. As the wave height increases, both the length and width of the sheltered zone decrease. As shown in the results, the length and width of the sheltered area decrease to varying degrees as wave height increases. Specifically, the sheltered area for waves with $H_0 = 0.12$ m is 29% smaller in length and 20% smaller in width relative to the case with the smallest wave height ($H_0 = 0.02$ m).

According to the derivation in Section 2 based on linear wave theory, the wave number inside and outside the waveguide governs how waves refract when they interact with the structure, independent of wave height. However, consistent with the wave patterns in Fig. 5, diffraction also plays a secondary role on the downstream wave fields, which is dependent on wave height. In the following analysis, wave fields under different wave height conditions are examined to assess how wave height affects the length of the sheltered zone.

Two representative cross-sections are selected for further analysis. The first is section B-B', located at the middle location of the waveguide, where wave transformation is primarily attributable to secondary refraction at the waveguide boundary. This section clearly illustrates the characteristics of wave refraction. The second is section C-C', situated in the wave height returning region behind the waveguide, which serves to analyze variations in the intensity of wave diffraction.

As shown in Fig. 17, the wave height along the B-B' profile shows little variation with increasing incident wave height. In contrast, along the C-C' profile, the wave height increases noticeably with incident wave height at $y/L_1 = 0$. Moreover, the zones of wave height increase at both lateral positions shift laterally, with the peak location deviating progressively from the waveguide centerline ($y/L_1 = 0$) as the incident wave height grows. It can thus be concluded that, under a constant wave period, wave height has a limited influence on wave refraction behind

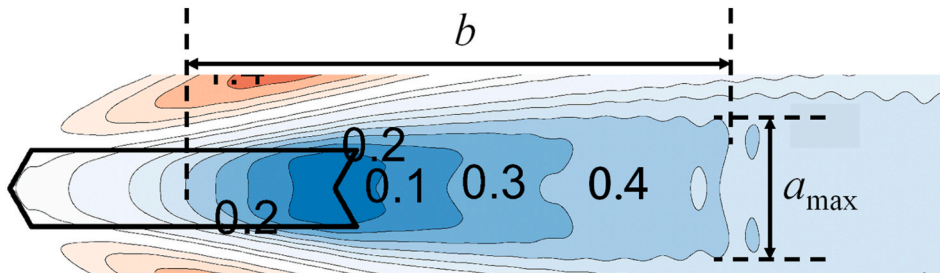


Fig. 12. Definition diagram of sheltered zone length b and maximum width $a_{\max}$.

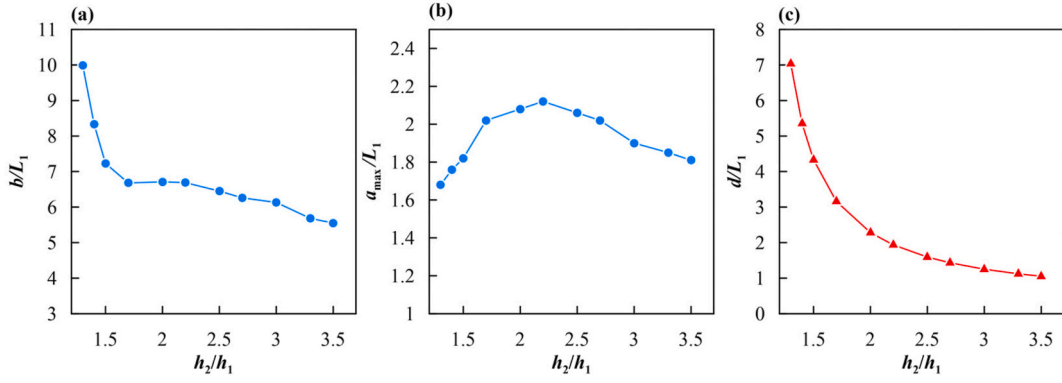


Fig. 13. Relationship between normalized sheltered zone length b/L_1 , normalized sheltered zone width $a_{\max}/L_1$, waveguide normalized length d/L_1 and depth ratios (h_2/h_1).

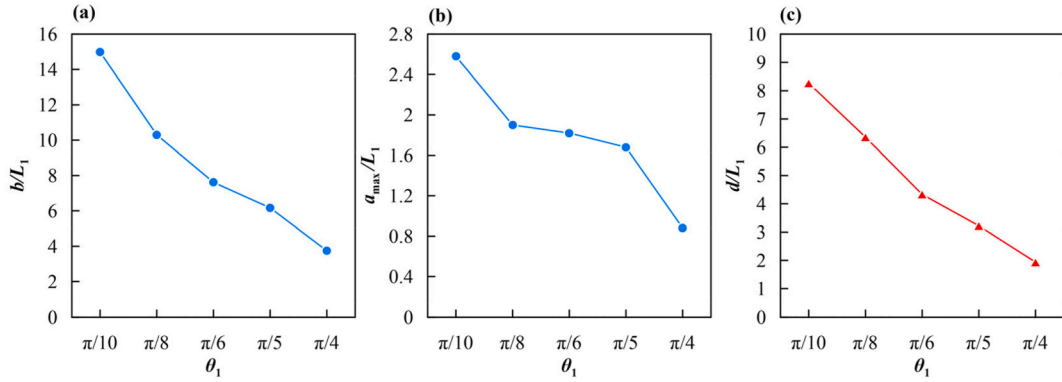


Fig. 14. Relationship between normalized sheltered zone length b/L_1 , normalized sheltered zone width $a_{\max}/L_1$, waveguide normalized length d/L_1 and waveguide front wing angle (θ_1).

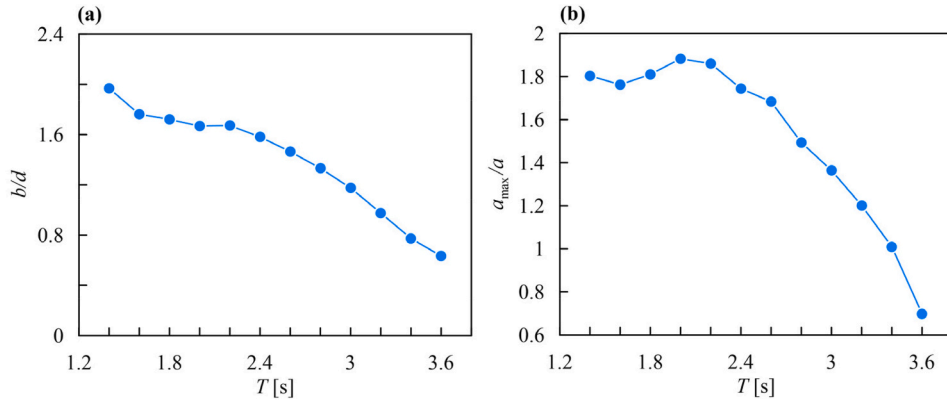


Fig. 15. The normalized length b/d of the sheltered zone and normalized sheltered zone width $a_{\max}/a$ behind waveguide trenches of the same size (based on design with $T = 2.0$ s incident waves, front-wing angle = 30° , $h_2/h_1 = 1.5$) versus the period of the incident waves.

the waveguide. However, larger incident wave heights enhance diffraction effects, which weakly modify the extent of the sheltered region, leading to a reduction in its effective length.

4.3. Optimization of waveguide design

The analysis reveals a strong relationship between the sheltered zone length (b) and waveguide geometry, where depth ratio (h_2/h_1) and front-wing angle (θ_1) jointly determine structural dimensions. Due to the resources required to construct a waveguide depends on how much of the bathymetry is altered, optimizing these parameters becomes critical for

cost-effective implementations. To quantify efficiency, we introduce a normalized sheltered area parameter R :

$$R = \frac{S \times h_1}{V} \quad (20)$$

where S denotes the sheltered area (defined as before, where $H/H_0 < 0.4$), V represents excavation volume, and h_1 is the original water depth. As shown in Fig. 3, viewed from above, the waveguide comprises two symmetrical parallelograms, each with a length of d and a height of $a/2$. In cross-section view, the waveguide exhibits a uniform depth of $h_2 - h_1$. Therefore, the excavated volume of the waveguide can be calculated as

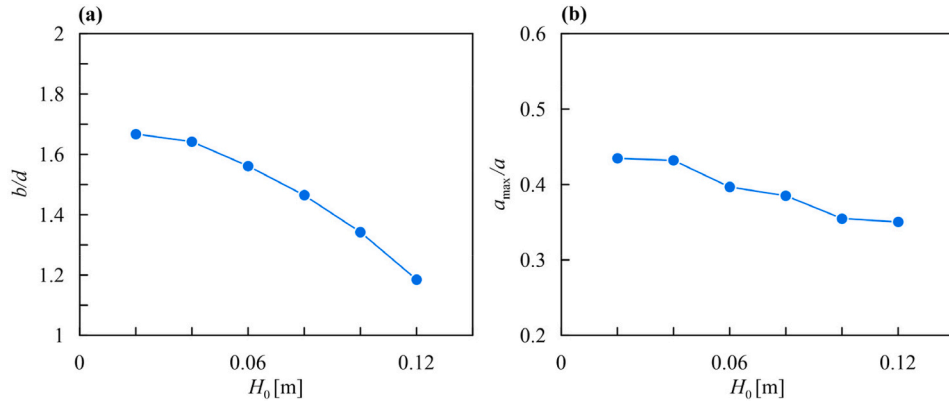


Fig. 16. The normalized length b/L_1 of the sheltered zone and normalized sheltered zone width $a_{\max}/L_1$ behind waveguide trenches of the same size (based on design with $T = 2.0$ s incident waves, front-wing angle $= 30^\circ$, $h_2/h_1 = 1.5$) versus the wave heights of the incident waves.

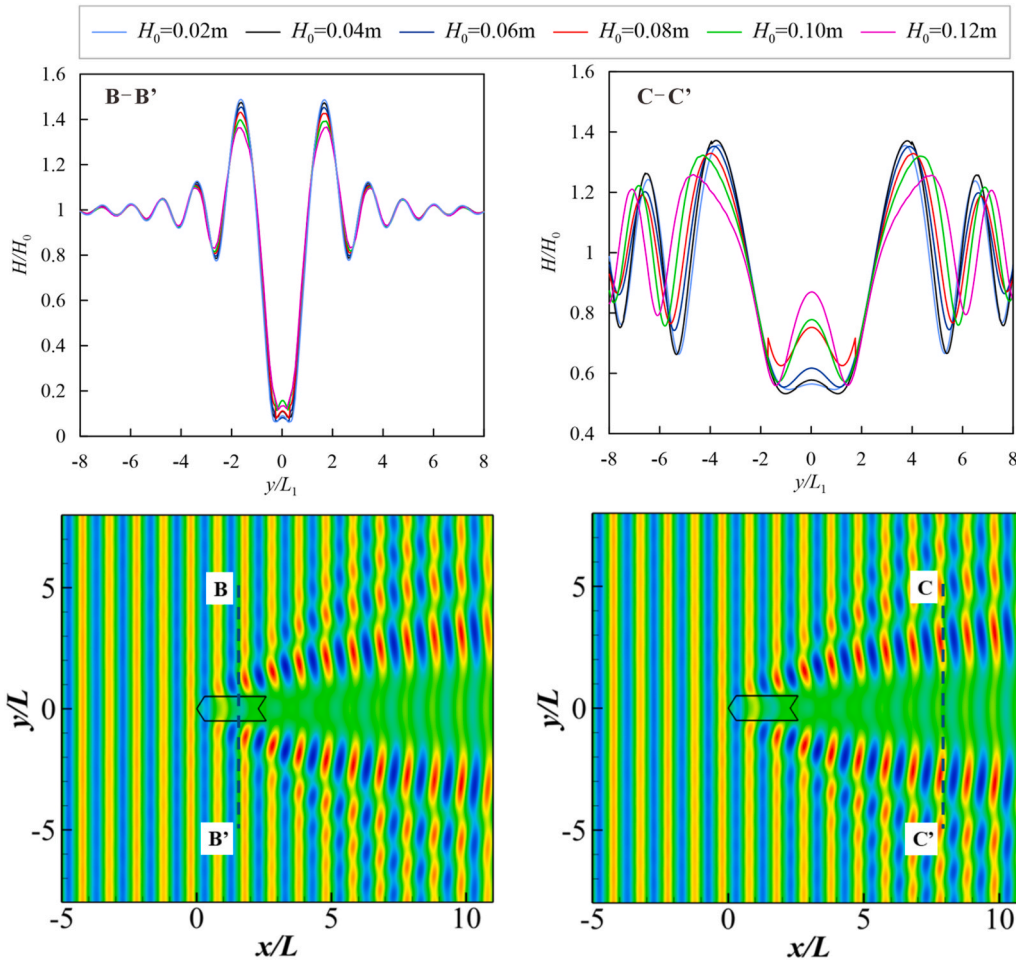


Fig. 17. The positions of the B-B' and C-C' cross-sections, along with the variation in dimensionless wave height (H/H_0) along the y-axis for these sections.

$V = [a/2 \times d \times (h_2 - h_1)] \times 2$. Given that the value of R represents the sheltered area generated per unit volume of excavation, a higher R value indicates that a larger sheltered area can be achieved for a given volume of excavated material in the waveguide trench unit, implying greater economic efficiency.

In the following analysis, we show that the ratio of the internal to external water depth, h_2/h_1 , and the angle between the front wing and the incident wave, θ_1 , represent two critical parameters in waveguide design due to their influence on R . For the analysis of the water depth

ratio h_2/h_1 , the following parameters were held constant based on the default values in Table 1: $h_1 = 0.4$ m, wave height $H_0 = 0.02$ m, waveguide width $a = 3.69$ m, and front-wing angle $\theta_1 = \pi/6$. For the analysis of the angle θ_1 between the incident wave and the front wing, the water depth ratio was fixed at $h_2/h_1 = 1.5$, with all other parameters consistent with those used in the water depth ratio analysis. To identify general principles, three incident waves with periods $T = 1.6$ s, 2.0 s, and 2.4 s, along with their corresponding waveguide designs, were analyzed.

Fig. 18(a) indicates that as h_2/h_1 increases from 1.1 to 3.5, a

declining trend is observed in each of the wave period cases. Maximum R occurs around $h_2/h_1 = 1.3$. Therefore, it may be concluded that, with the angle of the front-wing and the incident wave conditions remaining constant, a shallower excavation depth facilitates the formation of a larger sheltered area at the rear. Furthermore, it increases the sheltering zone per unit volume excavated, thereby enhancing economic efficiency.

Fig. 18(b) illustrates the relationship between the R value and the incident angle, θ_1 . Unlike the trend observed for the water depth ratio, the maximum R value occurs at different θ_1 for different wave periods. Specifically, the most economical design is achieved at $\theta_1 = \pi/4$ for $T = 1.6$ s, $\theta_1 = \pi/5$ for $T = 2.0$ s, and $\theta_1 = \pi/6$ for $T = 2.4$ s. This indicates that the design of the angle θ_1 between the front wing and the incidence wave is related to the incident wave period. For short-period incident waves, a larger θ_1 yields better economic performance, whereas for long-period waves, a smaller θ_1 provides superior economic efficiency.

5. Discussion & limitations

While this study demonstrates the theoretical feasibility and hydrodynamic performance of the butterfly-shaped waveguide, several limitations regarding geometric idealization, sea state complexity, and morphological stability should be acknowledged to guide future research and practical engineering applications.

5.1. Influence of boundary geometry

The numerical simulations in this study were conducted using a waveguide with vertical sidewalls (abrupt depth transitions). This geometric idealization was adopted to strictly align with the theoretical refraction framework derived from Snell's Law (Eq. (1)), which assumes a distinct interface between media of different phase velocities. Consequently, the results presented in Section 4 primarily reflect the wave steering capabilities driven by the refractive index contrast, isolating the fundamental refraction mechanism from other morphological factors. However, as noted in hydraulic engineering practice, vertical boundaries can induce flow separation, vortex shedding, and turbulence, potentially leading to local energy dissipation (Johnson and Davey, 1990). More critically, in practical coastal engineering applications, the construction of vertical sidewalls is often constrained by geotechnical stability, and the use of sloped or stepped sidewalls is typically recommended to ensure seabed stability and minimize sediment transport. Previous studies on submerged trenches (Smit et al., 2015; Thomson et al., 2007) indicate that while sidewall slopes influence the reflection coefficient, the fundamental refraction mechanism—driven by the phase velocity change due to depth difference—remains valid. It is anticipated that replacing vertical walls with sloped walls in real-world implementations would reduce boundary reflection, potentially allowing

greater wave energy transmission into the waveguide for steering. This suggests that the current results based on vertical walls may represent conservative estimates, and optimized sloped boundaries could maintain or even enhance the sheltering performance. Future research should focus on optimizing these sidewall slope gradients to balance structural stability with hydrodynamic efficiency.

5.2. Waveguide performance in realistic sea states

The simulations in this study were primarily conducted using regular waves to validate the waveguide function governed by Snell's Law. However, realistic sea states are characterized by random waves with directional spreading and broadband frequency spectra. Although full spectral simulations are beyond the scope of this study, the potential effects of these realistic factors can be inferred from the results presented in Section 4.

Regarding to wave directional spreading, in practical deployments, the central axis of the waveguide should be aligned with the principal wave direction. According to the design framework in Section 2, this arrangement establishes an optimal angle between the front wing and the principal wave, ensuring highly effective refraction and a strong sheltering performance for the primary wave components. However, a realistic directional sea state encompasses waves arriving from multiple off-axis directions. Based on the law of refraction, these non-principal wave components will interact with the front wing at incident angles that deviate from the optimal design angle. As demonstrated in Fig. 14, the waveguide's sheltering efficiency is affected by the incident angle (θ_1). Consequently, wave energy arriving from these oblique directions may undergo less effective refraction, likely resulting in a reduced overall sheltering effect.

Regarding broadband characteristics, Fig. 15 indicates that the sheltering performance is also period-dependent. Since the refractive index $n = C_1/C_2$ varies with frequency, a broadband wave field would experience frequency-dependent refraction angles. High-frequency components (short waves) might be refracted more sharply, while low-frequency components (long waves) might exhibit weaker steering. Therefore, in practical engineering applications, the waveguide design should be primarily optimized according to the peak period (T_p) of the target sea area. Nevertheless, based on the aforementioned derivations, the lower-frequency (long-wave) components within a broadband spectrum will experience smaller refraction angles, allowing more wave energy to exit from the back wing rather than being steered away. Consequently, these long-wave components may inevitably compromise the overall wave attenuation, resulting in an actual sheltered region in a realistic random sea state that is less extensive and less effective than the idealized conditions simulated using monochromatic waves.

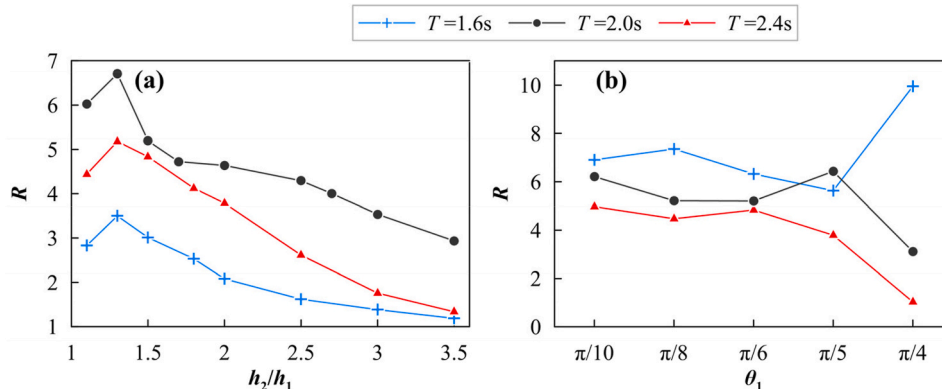


Fig. 18. Relationship between the value of R and the normalized excavation volume V^* and the ratio of different water depths inside and outside the waveguide structure.

5.3. Waveguide internal morphological stability

Finally, this study focused exclusively on the hydrodynamic performance of the waveguide structure without considering seabed evolution. Since the waveguide functionality relies on a specific trench geometry to maintain the refractive index contrast, sediment transport and backfilling could significantly affect its long-term functionality. In high-energy coastal environments, the trench could act as a sediment trap, altering the water depth ratio (h_2/h_1) over time. Subsequent studies should incorporate sediment transport models to assess the morphological stability of the trench, predict backfilling rates, and determine the necessary maintenance dredging requirements to ensure the longevity of the system.

6. Conclusions

In this study, a novel waveguide is proposed to exploit the wave refraction mechanism induced by depth variations. The following conclusions can be drawn from numerical simulations:

- (1) In accordance with the design of the waveguide, the amplitude of the wave diminishes in the rear section of the waveguide. However, as waves pass through the waveguide, secondary refraction at its sides causes them to interact with the incident waves, forming a fan-shaped region in which the wave height increases. A thorough examination of the fan-shape region has been undertaken, and the results of this examination have indicated that as the water depth inside the waveguide increases, the angle of the fan-shaped region also increases. Moreover, the maximum angle between the fan-shaped region and the incident wave direction is larger than the theoretically derived angle of secondary refraction, and the difference between the two angles increases as the water depth increases.
- (2) The waveguide designed based on the incident wave achieved consistent wave attenuation effects within the shielded zone ($K = H/H_0 < 0.4$ region) in all tested configurations, with a maximum attenuation rate exceeding 90%; Increasing the water depth ratio reduces the guide structure length d/L_1 and the sheltering length b/L_1 , but when $h_2/h_1 > 1.7$, the marginal gain of b/L_1 decreases sharply; a higher water depth ratio ($h_2/h_1 \geq 2.0$) causes the low wave height region ($K < 0.1$) to shift to both sides while expanding the lateral influence range. Under the condition of a fixed water depth ratio $h_2/h_1 = 1.5$, increasing the front-wing angle will cause d/L_1 to decrease linearly by 58% and b/L_1 to decrease by 43%.
- (3) Numerical simulations were used to evaluate the shelter effect of a fixed-size waveguide on different incident wave parameters. The results showed that the sheltering effect of the waveguide is sensitive to the incident wave period. When the waveguide is designed for an incident wave period of T_d , comparing normalized sheltered area lengths (b/L_1) in different incident wave periods, the length for a wave period of $1.8T_d$ is about 90% of that for $0.7T_d$. In addition to the wave period, incident wave height also influences the sheltering performance. With the incident wave period fixed at T_d , increasing the wave height to six times the smallest incident wave height H_0 leads to a 29% reduction in breakwater length and a 20% reduction in width compared to the H_0 condition. Analysis of wave heights at different positions reveals that increased wave heights lead to greater diffraction behind the waveguide, thereby affecting the sheltering effectiveness.
- (4) This paper further explores the economic efficiency of trench excavation for waveguides. Analysis of the sheltered area per unit excavation volume demonstrates that a water depth ratio (h_2/h_1) around 1.3 provides the best economic efficiency in waveguide design. The economic performance of the front-wing angle θ_1 ,

however, is period-dependent. A smaller θ_1 is more advantageous for long-period waves, while a larger θ_1 yields greater benefits for short-period waves.

CRedit authorship contribution statement

Bowen Guan: Writing – original draft, Validation, Methodology, Conceptualization. **Yuxiang Ma:** Writing – review & editing, Methodology, Funding acquisition, Conceptualization. **Ryan Lowe:** Writing – review & editing, Validation, Supervision. **Congfang Ai:** Validation, Software.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This research is supported financially by National Key Research and Development Program of China (Grand No. 2024YFD2400103).

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